MUHAMMAD BINTANG INDRA HIDAYAT

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Software Developer with 1+ year of professional experience building systems for healthcare and retail, including a Hospital Management System used across 5+ hospitals in Aceh and Corepotek, a pharmacy platform serving five clients. Experienced in applied AI, including machine learning, computer vision, NLP, LLM integration, prompt engineering, and RAG. Recipient of five national awards in data science, AI, and digital innovation.

Work Experiences

PT Paripurna Sentosa Digital - Aceh Barat, Indonesia Oct 2025 - Present

Software Developer

- Develop SIMRS modules with Vue.js, PHP (CodeIgniter) REST APIs, and MySQL, supporting operational workflows across 5+ hospitals in Aceh.
- Resolve 5-10 user-reported issues daily through troubleshooting, bug fixing, and system maintenance.
- Built an LLM-powered Telegram bot that converts staff complaints into structured IT tickets and automates assignment and status tracking.
- Contributed to Corepotek, a pharmacy management system launched in September 2026 and now in use by five clients across Nagan Raya and Aceh Barat.

Gutech - Banda Aceh, Indonesia (Remote) Aug 2026 - Present

Freelance Full-Stack Web Developer

- Built Metha Learn (methalearn.id), an online learning platform, with a Go REST API (JWT, Redis, PostgreSQL), a Next.js front end, and Docker/GitHub Actions deployment.

Physics Department, USK - Banda Aceh, Indonesia Jun 2025 - Sep 2025

Full-Stack Web Developer

- Built a Laravel and MySQL laboratory management platform for equipment lending, visitor scheduling, and service requests, with 59 automated feature tests across six modules.

Education

Universitas Syiah Kuala - Banda Aceh Jun 2022 - May 2026

Bachelor of Informatics, 3.82/4.00, Cum Laude

- Relevant Coursework: Software Engineering, Artificial Intelligence, Machine Learning, Database Systems, Object-Oriented Programming, Data Analysis & Visualization, Information Systems.

Projects

Hybrid YOLO-CNN with Hierarchical Classification (1st Place, LOGIKA UI 2025 Data Science) **Github Link**

Skills: Deep Learning, Computer Vision, Object Detection, Model Fine-Tuning, Data Preprocessing, Model Evaluation

- Built a hierarchical classification pipeline that combines YOLOv12 object detection with per-category CNN classifiers to classify traditional Indonesian houses from photos, reaching 0.955 macro F1 versus 0.901 (direct CNN) and 0.664 (YOLO alone).
- Fine-tuned pretrained CNN architectures (VGG16, MobileNetV2, ResNet50, EfficientNetB0) per object category, and cleaned the dataset by hand (2,196 images, 262 mislabeled or irrelevant removed) with upsampling for imbalanced classes.

LawGraph ID: Graph-RAG for Indonesian Regulations **Github Link**

Skills: NLP, Model Fine-Tuning, RAG, Vector/Semantic Search, Knowledge Graph, Data Pipeline

- Built a question answering system over 63 Indonesian regulations (2,409 articles), with an end-to-end data pipeline from scraping source PDFs to a Neo4j knowledge graph, including automated data quality checks.
- Developed a LangGraph agent served through a FastAPI backend that routes between Text2Cypher graph queries and semantic search with graph expansion, with read-only query execution and automatic fallback.
- Fine-tuned an Indo-LegalBERT embedding model on 1,299 question-article pairs, evaluated it with recall@k and MRR (0.71 recall@14, 0.60 MRR on a 60-question benchmark), and published it on Hugging Face.

CIRVIA (Bronze Medal, LIDM 2025) **Github Link | Website**

Skills: Next.js, Supabase, Computer Vision (MediaPipe), Generative AI, Adaptive Learning

- Developed an AI-powered physics learning platform combining virtual circuit simulation, real-time hand gesture recognition (MediaPipe, OpenCV), and adaptive learning.
- Integrated a generative AI feedback feature (OpenAI, Gemini) that analyzes each student's completed work and delivers tailored feedback, supported by a teacher analytics dashboard for monitoring student progress.

SIMRS IT Support Bot **Github Link**

Skills: Python, LLM Integration, Prompt Engineering, Telegram Bot API, Google Sheets API

- Built a Telegram bot that uses an LLM to convert free-text staff complaints into structured IT tickets (category, priority, description), enforcing JSON output with safe defaults when the model deviates.
- Automated the ticket lifecycle: Google Sheets logging, IT group routing, chat-based assignment, and reporter notifications.

WanTuTri AI (Top 10 Finalist, Alibaba Cloud AI x Creativity Hackathon 2025) **Github Link | Website**

Skills: Next.js, Prompt Engineering, LLM Integration, Generative AI, Workflow Automation

- Built an AI platform that turns a product idea into a promotional video through an automated pipeline of script generation, scene visuals, and video production.
- Designed multilingual (EN/ID/ZH) prompts for Alibaba Cloud Qwen and WanAI models to generate structured scripts and scene visuals.

Skills

- **Core Competencies:** Full-Stack Development, RESTful API Development, Database Design, Automated Testing, Troubleshooting & Debugging, Machine Learning, Computer Vision, NLP, Model Fine-Tuning & Evaluation, LLM Integration, RAG, Prompt Engineering.
- **AI & Data Tools:** Python, TensorFlow/Keras, PyTorch, scikit-learn, Ultralytics YOLO, OpenCV, MediaPipe, Hugging Face, LangGraph, LangChain, Neo4j, OpenAI, Gemini, Qwen, Groq, Ollama.
- **Programming Languages:** Python, TypeScript, JavaScript, Go, PHP, SQL, Java, Kotlin, Dart.
- **Frameworks & Libraries:** Next.js, React.js, Vue.js, Node.js, Express.js, NestJS, Laravel, CodeIgniter, GoFiber, GORM, FastAPI, Flutter.
- **Databases & Tools:** MySQL, PostgreSQL, Redis, MongoDB, Supabase, Neo4j, Docker, Docker Compose, Git, GitHub Actions.
- **Certification:** Artificial Intelligence - Badan Ekraf Digital Talent (2026), SAP Master Data Governance - SAP (2026), Data Analyst SAP Analytics Cloud - SAP (2026)

Achievements

- **1st Place LOGIKA UI 2025 Data Science Competition Universitas Indonesia (2025):** Outperformed 94 competing teams to claim 1st place among 10 national finalists.
- **Bronze Medal LIDM (Lomba Inovasi Digital Mahasiswa) by Kemdiktisaintek (2025):** Awarded Bronze Medal after placing among the top 3 teams from 20 finalists, selected from over 200 submissions nationwide.
- **2nd Place Hack4Health Hackathon (Sandbox 2.0) (2025):** Secured 2nd place among 10 finalist teams by developing innovative technology solutions for healthcare challenges.
- **3rd Place E-Commerce Design Category, The 11th UTU Awards (2025):** Ranked 3rd among more than 50 submitted projects in the national E-Commerce Design competition.
- **Top 10 Finalist AI x Creativity Hackathon, Alibaba Cloud (2025):** Selected as one of the top 10 finalists in a highly competitive national hybrid hackathon.

Organisational Experience

BEM Faculty of Mathematics and Natural Sciences Feb 2024 - Feb 2025

Human Resource Development Staff

- Led the organizing committee of 15+ members in planning and executing a faculty-level sports tournament, ensuring smooth coordination from planning through event implementation.
- Coordinated the faculty futsal team of 20+ members, overseeing recruitment, team communication, and participation in inter-faculty competitions.